

40Gbps Wi-Fi technology

Researchers from the Fraunhofer Institute for Applied Solid State Physics and the Karlsruhe Institute of Technology recently demonstrated a wireless technology capable of data uplink speeds of 40 Gbps at 240GHz frequency. This is roughly 137 times faster than the current 802.11n (Wi-Fi) standard, which supports speeds of up to 300 Mbps. In layman terms, it would help you transfer several high-definition movies within a second! The technology is also capable of transmitting across a large range of 1 km. The team showed the results by transmitting data from one skyscraper to another. Part of the European Millilink project, the research is funded by the Federal Ministry of Education and Research. Firms like Siemens are also assisting in the research, hoping to commercialise the design in the coming years.

Inside: Wireless components fit into a single, small chip, making the technology suitable for use in smartphones, laptops and other mobile devices. The active transmitter and receiver design fits on a $4 \times 1.5\text{mm}^2$ integrated circuit (IC), designed using special high-electron-mobility transistors.

The researchers managed to make the chip so tiny by using very high frequencies in the 200-280GHz band as compared to today's 2.4 or 5GHz Wi-Fi. Higher frequencies mean smaller components, since a shorter wavelength can be picked up by a smaller antenna. Moreover, the number of bits that can travel over airwaves is inversely proportional to the wavelength. So a shorter wavelength means this high-frequency technology can push much more data too. The antenna also has excellent signal fidelity.

One of the team members explained in a press release that the new design showed low attenuation in this frequency range, which enabled broadband directional radio links. This makes the radio link easier to install than free-space optical systems for data transmission. It is also more robust in poor weather conditions.

Web: http://www.kit.edu/visit/pi_2013_12950.php



Adaptable and flexible 'soft-modem'

At the heart of NVIDIA's Tegra 4i mobile processor for mainstream smartphones is its integrated i500 LTE modem that is capable of working through 100-150 Mbps of LTE data. NVIDIA's i500 modem is based on the software-defined radio technology developed by Icera, which NVIDIA recently acquired.

The i500 is highly adaptable and flexible, making it easy for carriers and OEMs to update modem features and to take advantage of performance-improving algorithmic innovations in the field through over-the-air updates. The modem is also tiny—just about 40 per cent the size of conventional LTE models—as it is designed with general-purpose deep execution processors. It is also multi-mode—it delivers 4G LTE Advanced and is backward compatible with LTE Cat 3, 3G and 2G too.

Inside: In conventional modems, most features are implemented in silicon, which results in a large die size and limited flexibility. The i500 overcomes these issues with software-defined radio technology, which not only supports the latest wireless protocols such as DC-HSPA+ and LTE but also provides the flexibility of being fully programmable to support next-generation interfaces and standards. The i500 is considered by many experts as the most software-based modem in the market, compared to similar claims from companies like Qualcomm.

In the case of the i500, the entire digital baseband is just one big platform, on top of which the software can be run. There is an external transceiver for down-conversion but beyond that it is pure software. The i500 uses TSMC's 28nm high-performance, high-K metal gate process. The i500 features a small footprint and a high-performance process, yet conserves power by turning off blocks that are not in use—compared to a larger system-on-chip (SoC) with lower leakage. As a result, NVIDIA's i500 claims a 40 per cent smaller die than Qualcomm's MDM9x15.

Web: <http://www.nvidia.com/object/i500-cellular-modems-products.html>



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